Sixty-Four Modes and Expanding Radiation-to-Matter Conversion in WRRA

MCC 3.0 Item 6 reinforcement, v0.3

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Verified inputs

We retain the v0.2 quantized 128-cell profile, address-generated background and patched CAMB 2.0.4 source. The previous 576 checks belong to the previous audit and are not counted again. Known observations are legitimate disclosed construction/calibration inputs; outputs remain distinguished from inputs.

We extend $k_n = 0.05n$ Mpc^{-1} to $n = 1, \dots, 64$ and calculate **all 5,999 integer multipoles** from 2 through 6000. These are new Boltzmann evolutions, not extrapolations of the seven cached modes. The outputs use linear T/E/lensing transfers; nonlinear structure and a new observational likelihood are not executed.

New dynamical choices are $a_i = 10^{-10}$, a proper-time conversion pulse lasting one initial Hubble-time unit, and $\Gamma = \ln 2$ in that unit. The existing quantized seed modulates the proper rate as $\Gamma_j = \Gamma(1+S_j^{(0)})$. This is a declared continuum fluid-rate closure, not a calculated particle cross section or a finite-quantum emission-time law.

WRRA transformation

Complete mode and multipole coverage

A single dense Bessel table exceeded memory. We instead calculate eight shifted paired-parity multipole grids per initial basis, and merge their actual transfer values. Their union contains every integer multipole. Overlap agreement and complete coverage are checked. **No missing-multipole interpolation or wavenumber extrapolation is used in the final arrays.** Both D-isocurvature and A-adiabatic transfers are evolved.

For the finite 128-cell Fourier transform, mode variances are $v_n = 2|\widehat{S}_n|^2$ for $n < 64$ and $v_{64} = |\widehat{S}_{64}|^2$. Under the retained real Nyquist cosine and 128 discrete translations, the Nyquist mode contributes to even multipoles only. This is an adopted finite-grid readout property, not a universal cosmic parity law. An additional Nyquist sine state would define a different preparation.

Expansion, conversion and pressure work

The earlier half-conversion reference is interpreted as half conversion of a small convertible radiation subreservoir, not half of all cosmic radiation. Its calibrated fraction is $\beta = 7.62452468 \times 10^{-7}$, fixed by the adopted observed CDM/radiation background ratio. This is a disclosed calibration input.

With $8\pi G/3 = 1$, initially $a = H = 1$, define $N = \ln a$, $C = \rho_c a^4$, $U = 1 - \beta$, and $M = \rho_m a^3$. In each cell's proper time,

$$\dot{N} = H, \quad \dot{C} = -\Gamma_j C, \quad \dot{M} = \Gamma_j C e^{-N}, \quad \dot{H} = -\tfrac{1}{2}(4\rho_r + 3\rho_m),$$

$$\rho_r = (U+C)e^{-4N}, \quad \rho_m = Me^{-3N}, \quad \dot{W} = H(U+C)e^{-N}, \quad E = (U+C)e^{-N}+M, \quad E+W = 1.$$

H evolves independently, so $H^2 = \rho_r + \rho_m$ is a nontrivial constraint check rather than an overwritten identity. Exchange terms enter the two components with opposite signs. This implements $Q_m^\mu = -Q_r^\mu$

in a common comoving velocity at leading gradient order. These are local patches of one universe, not multiple physical universes. Local proper-time clocks do not introduce a universal absolute simultaneity. Unconverted radiation redshifts and does pressure work; its energy alone is not incorrectly held constant.

Dynamical entropy and curvature enter actual CMB bases

Tangent ODEs propagate the small rate modulation without subtracting nearly equal full solutions. After exchange ends,

$$\zeta_r = \frac{\partial_\xi C}{4(U+C)}, \quad \zeta_m = \frac{\partial_\xi M}{3M}, \quad S = 3(\zeta_m - \zeta_r).$$

The selected closure gives $S = 0.723567385056 S^{(0)}$ and $\zeta_r = -6.60614975286 \times 10^{-8} S^{(0)}$. These are correlated amplitudes from the same preparation. Actual CAMB initial callbacks establish the initial-vector mapping $(-\zeta_r, S, 0, 0, 0)$. We therefore combine $S\Delta_D - \zeta_r \Delta_A$ before forming covariance, retaining the cross term. The generated curvature is not silently deleted.

Outputs

Audit or quantity	Result
Projection/coverage checks	96 passed
Dynamics checks	15 passed
Initial-basis checks	4 passed
Separate precision/slicing/Nyquist controls	8 passed
Untransferred variance within the declared 128-cell model	0%
Maximum normalized old-seven transfer discrepancy	9.5725592e-05
Maximum standardized direct-phase covariance discrepancy	6.3596178e-16
Energy plus pressure-work error	below 2.0×10^{-14}
Relative Friedmann constraint error	below 2.8×10^{-12}

Projected output	Static 64-mode preparation	Dynamical 64-mode preparation
T RMS [μK]	0.00172883455993	0.0012509283125
E RMS [μK]	6.39105354627e-05	4.62436378837e-05
Deflection RMS [arcsec]	0.00232005986269	0.00167871840042

The newly transferred 57 modes contribute 1.6121617e-05%, 9.1445155e-05% and 0.048377115% of static projected temperature, E-polarization and deflection variance, respectively. These differ from the initial entropy-variance fraction of 10.897%; the CMB transfer weights are now actually applied.

Matching occurs near $a = 4.58258 \times 10^{-10}$, with maximum $k/(aH) = 6.79764 \times 10^{-4}$. Its square, 4.62079×10^{-7}, is a gradient-expansion parameter, **not a rigorous bound on all omitted matching errors**. The calibrated proper-time unit is about 0.477 seconds, conditional on the chosen starting scale and simple effective-radiation history; it is not a detailed MeV thermal-history chronology.

At 471 actually evolved multipoles, raising precision from 2.5 to 3.0 changes the power L2 norm by 0.232% for T, 0.0156% for E and 0.0973% for the lensing potential. Printed digits identify reproducible numerical outputs, not physical precision or detection. An independent nonlinear uniform-density-slice calculation agrees with the tangent curvature response to relative 1.59×10^{-6}.

Refutation conditions

Missing modes/multipoles, contradictory overlaps or Nyquist double counting reject the transfer implementation. Fourier/covariance failure, failed direct-phase checks or failure to recover old modes requires repair. Negative densities, failed energy-work/Friedmann constraints or nonconserved post-exchange component curvatures reject the dynamical branch. The separate-universe approximation is not used if the required small-gradient condition fails. Changed pulse, reservoir or starting assumptions define a separately recorded branch. A complete fixed physical model must pass its specified observational comparisons.

Assessment and remaining scope

Modes 8-64 are transferred, and an expanding conversion closure with pressure work and curvature response is implemented. Dynamically generated entropy and radiation curvature are actually inserted into CMB initial bases. The new fluid closure and calibration freedoms are disclosed.

Full finite-gradient metric/momentum/free-streaming-neutrino matching, particle species and rates, finite-quantum emission timing and recording costs, detailed thermal history, nonlinear lensing and a new likelihood remain OPEN. CAMB's regular baryon/neutrino completion is adopted; the two-fluid model has not computed their production reactions. Thus Item 6 is partially implemented through leading-gradient dynamics, not completely closed. The small new component does not replace the entire observed CMB.

Reproduction and sources

DOI: 10.5281/zenodo.23489547. Code, inputs and computed projection data: GitHub.

Run `run_all.py`, `dynamics.py`, `check_basis.py`, then `analyze.py`. Sources, patches, runtime, inputs, arrays and hashes are included. The failed full-table attempt is distinguished from the successful split-grid evolutions.

- Previous v0.2: published study.
- CAMB official model and initial-basis documentation.
- Malik & Wands, Cosmological perturbations.
- Park, Equations of General Relativistic Radiation Hydrodynamics from Tensor Formalism.

External equations are disclosed construction and verification material. Only the internally executed scope is called integrated.

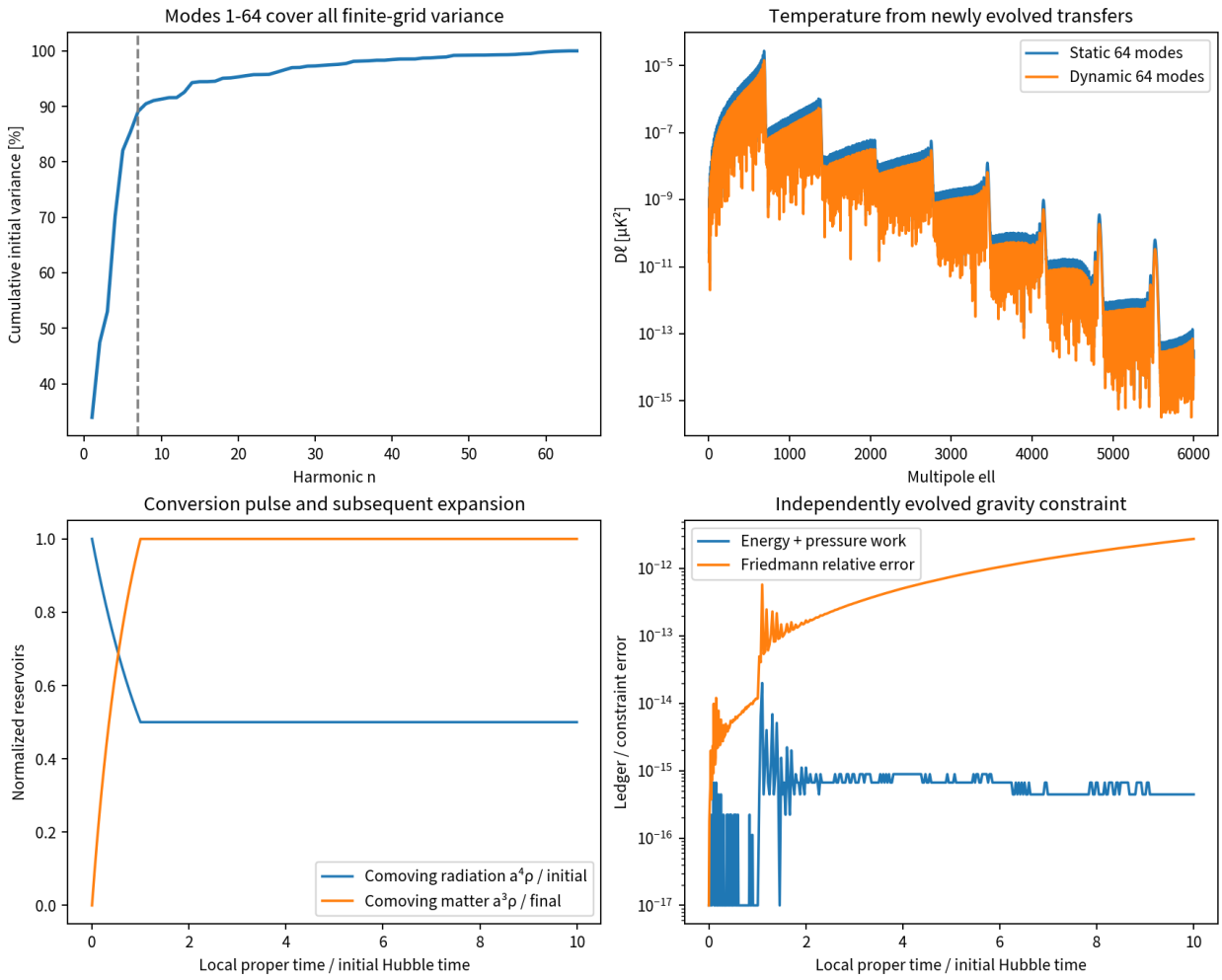


Figure 1: Full finite-mode variance coverage, CMB projection and expanding-conversion conservation diagnostics.